Definitions and Assumptions

Project: Gross margin decline, Q1 → Q3 2026 · **Version:** v1.0 · **Author:** Ogunlusi Teniola Elizabeth, Data Analyst · **Date:** 17 October 2026

This document fixes the meaning of every term and the basis of every rule used in the analysis, so a reader can see what was defined by Finance, what was proven from the data, and what is my own judgement.

Source key. F = confirmed by Finance (answers relayed to me) · **D** = proven from the data · **A** = my assumption or decision · **I** = inferred (consistent with the data, not stated by anyone)

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1. Purpose and scope

- **Brief received:** *"Margins feel off lately. Look into it."* The scope, definition of "done" and report outline were agreed with the requester: a version 1 report with KPI cards, a trend, a breakdown, and possibly a drill-through page, matching Finance's numbers.
- **What this analysis covers:** gross margin for orders dated 1 January to 28 September 2026 (1,890 orders; 4,712 order lines after cleaning), compared between Q1 and Q3.
- **What it is for:** the commercial committee's decisions on the price list, discount rules, and which products and customer segments to push.
- **Not covered:** contribution margin (freight, duty, warehousing), VAT treatment, returns or credit notes (the data has none of these), and any period after September 2026.

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2. Definitions confirmed by Finance

Finance's wording is quoted where it matters. Where each rule is applied is shown in the last column.

#	Item	Definition	Src	Applied in
2.1	**Gross margin %**	*(Net revenue − COGS) ÷ Net revenue*	F	`03`, `04`, `05`

2.2	**Net revenue**	Revenue **after discounts**, **excluding VAT**. In the data: kg × unit price (NGN) × (1 – discount %)	F	03
2.3	**COGS**	*kg sold × the supplier cost in USD per kg (the cost effective on the order date) × that month's FX rate from the fx_rates table*	F	03
2.4	**Costs excluded**	*Freight, duty, and warehousing aren't in this data or in Finance's number.* Gross margin here is therefore **not** a contribution margin	F	all
2.5	**Blank currency**	*Blank means NGN. The field was added later and was never mandatory.*	F	02
2.6	**USD-tagged lines**	*Lines tagged USD really are USD-priced. Convert them at that month's rate.*	F	03
2.7	**Duplicate lines**	*IT confirmed a sync bug that double-posted some lines, and 94 sounds right.* Rule: *exact copies (same order, product, qty, price, discount, and currency, with only line_id different) get one copy kept and the rest removed. Anything that isn't an exact copy stays. Log the count and the value you removed.*	F	02, 05
2.8	**Price list unit**	*The price list is per kg, always.*	F	03
2.9	**Bag-product quantities**	*The warehouse changed how bag products are recorded at some point, and I don't know when or how. Prove it from the data, and don't take my word for it.* Finance could not confirm the quantity unit; I proved it from the data (item 3.4)	F / D	03, 05
2.10	**Control totals**	Q1 2026: **₦5,191.2M** net revenue, **33.3%** GM. Q3 2026: **₦3,825.6M** net revenue, **19.6%** GM, on the same definitions, cleaned	F	05
2.11	**Reconciliation tolerance**	Within about **0.5% on revenue** and **0.2 points on GM %**. *If you're off, find your own gap first, then bring it to me.*	F	05
		Q1 2026 (January to March): the budget was built on it and it is the margin the board		

2.12	**Reference period**	expects. Do not only compare Q1 with Q3: show month by month when the margin started to slip, and say if Q1 is not representative and what to use instead	F	`04`, memo
2.13	**What the committee wants**	What changed, how much margin it cost in naira, and a recommendation with the likely impact of each option; no pile of charts. State uncertainty plainly (*"I'm 80% sure"*) and show what would be checked next	F	memo

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3. Rules proven from the data

#	Rule	Evidence	Src
3.1	**Slash dates are dd/mm/yyyy** (day first)	Of the 177 slash-format dates, 88 have a day above 12 and **none** has a month above 12	D
3.2	**`order_lines` was loaded twice** in my environment (9,612 rows = 2 × 4,806) and the extra copies are removed by the de-duplication rule	4,806 distinct `line_id` s in both source files	D
3.3	**Costs change only for two products**	22 cost rows = 20 products + 2 increases: P03 and P04 both +9.0% on 2026-06-01	D
3.4	**Bag products (P03 Potassium Sorbate, P16 Sodium Bicarbonate) are recorded in bags from 1 May 2026**, so kg = qty × 25 from that date	Before 1 May every quantity is a multiple of 25 (25 to 3,000); from 2 May the quantities are 1, 2, 4, 8, 10, 20, 40, 80, 120, none a multiple of 25; zero overlap. Last kg-style line 28 Apr, first bag-style line 2 May. Without the rule Q3 revenue is 6.8% short of Finance; with it, exact	D
3.5	**Discounts are whole-number percentages**	Values present: 0, 2, 3, 4, 5, then 10, 11, 12, 13; and the reconciliation ties exactly	D / I
3.6	**Deep discounts begin in July**	350 lines at 10–13%, none before 1 July	D

4. Time and grouping definitions

Time

- **Order date** drives everything (not delivery or invoice date). **Month** = the calendar month of the order date, and it selects the FX rate.
- **Q1** = 1 Jan–31 Mar · **Q2** = 1 Apr–30 Jun · **Q3** = 1 Jul–30 Sep 2026. The last order in the data is dated 28 September.
- **FX:** one rate per month, applied to every order in that month. Rates run from ₦1,480 per USD (January) to ₦1,720 (September).

Customer groups (assigned in `03_margin_model.sql` ; checked in this order)

Group	Rule	Customers
Distributor	`customers.segment = 'Distributor'`	C009, C018, C027, C036, C045, C054 (6)
Deep-discount	Not a distributor, and at least one order line with a discount of **10% or more**	C007, C019, C033, C061, C062, C063: four Bakery, two Confectionery (6)
All other	Everyone else	the rest

The 10% threshold sits at a natural break: discount values jump from 5% to 10%, with nothing between.

5. Metric glossary

Term	Meaning
kg	Quantity in kilograms (qty, or qty × 25 for bag products on or after 1 May)
Gross revenue	kg × unit price in NGN, before discount
Discount (₦)	Gross revenue × discount %
Net revenue	Gross revenue – discount (Finance's revenue figure)
COGS	kg × USD cost per kg (on the order date) × that month's FX

Gross profit (GM, ₦)	Net revenue – COGS
GM %	Gross profit ÷ net revenue
GM % ex-distributors	GM % excluding the Distributor group

Margin bridge effects (Q3 versus Q1, per product within each customer group; P = gross price per kg, d = discount rate, u = USD cost per kg, f = FX rate, V = kg; subscript 0 = Q1, 1 = Q3)

Effect	Formula	Meaning
Volume / mix	$(V_1 – V_0) \times$ Q1 unit margin	More or fewer kilos, and a shift between higher- and lower-margin products
NGN list price	$V_1 \times (P_1 – P_0) \times (1 – d_0)$	Change in naira price per kg
Discounts	$–V_1 \times P_1 \times (d_1 – d_0)$	Change in the discount rate
Supplier cost	$–V_1 \times (u_1 – u_0) \times f_0$	Change in the USD supplier cost, at Q1 FX
FX	$–V_1 \times u_1 \times (f_1 – f_0)$	Change in the exchange rate on the USD cost

The five effects add up exactly to the change in gross profit (check V12 in `05_validation_checks.sql`). The attribution is sequential: rate effects are measured at Q3 volumes and volume/mix at Q1 unit margins, so a different ordering would move amounts between rows but not the totals.

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6. Assumptions register

ID	Assumption	Src	If it is wrong	Status
A01	Blank currency means NGN (107 lines)	F	Those lines would be mispriced	Confirmed by Finance; the prices are NGN-sized
A02	USD-tagged lines are genuinely USD-priced	F	Unconverted, Q1 revenue is 0.9% low and Q1 GM	Confirmed by Finance; effect estimated

	(27 lines)		0.6 points low	
A03	Unit prices are already excluding VAT	I	Revenue overstated by the VAT rate	The data has no VAT field; the exact tie to Finance's ex-VAT figures supports it
A04	`discount_pct` is a percentage (3 means 3%)	I	Revenue wrong by a factor of 100	The exact tie to Finance supports it
A05	One FX rate per month applies to the whole month	A	Intra-month moves are missed	Rates table supplied for January to September only
A06	Cost used is the latest `effective_date` on or before the order date	F	COGS misstated for P03 and P04 after 1 June	Applied as defined by Finance
A07	Unit price is per kg for every product, including bag products	F	Bag product revenue misstated	Confirmed by Finance
A08	Bag quantities are in bags from 1 May 2026	D	Without the rule, Q3 revenue is 6.8% short	**Proven from data; not confirmed by the warehouse** (open item)
A09	Only exact copies are duplicates; the lowest `line_id` is kept	F / A	Over- or under-removal	Rule from Finance; the choice of lowest `line_id` has no numeric effect because copies are identical
A10	Lines sharing an order and product but not exact copies are genuine	F	Revenue overstated	Kept per Finance's rule; 26 distributor lines in Q3 remain unexplained (see README §7)
A11	"Distributor" is as labelled in `customers.segment`	A	Group totals change	Six customers carry the label
A12	Deep-discount = any line of 10% or more	A	Group membership changes	Natural break in the data (nothing between 5% and 10%)

A13	Blank rep names are labelled "Unassigned" (88 orders)	A	Rep-level views only	No effect on margin totals
A14	Q1 is the reference period	F	A different baseline changes the size of the drop	Set by Finance; limits discussed in the decision memo
A15	Q2 is calculated the same way but has no Finance control total	A	Q2 may carry undetected error	Open: not independently reconciled
A16	Distributor behaviour change is real, not another data error	A	The ₦572M finding would be overstated	Judged ~85% sure; IT confirmation requested
A17	Option impacts assume volume holds or falls 5–10% at the Q3 run-rate	A	Impacts over- or understated	Indicative; **not produced by the SQL files**
A18	The database is SQLite, run in DBeaver	A	Syntax differences on other engines	Scripts ran without error in my DBeaver session

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7. Change log

Version	Date	Change
v1.0	17 October 2026	First issue